

Credit Risk Analytics for Smarter Lending Decisions

Executive Insights from
32,000+ Loan Applications

Why This Matters

Default Risk Is Real

Default Risk Is Costing Us Millions

1 in 5 loans fail — that's over \$68 million in exposure.

Our Portfolio Needs Smarter Protection

Current lending patterns expose us to avoidable risk.

Smarter Lending Means Fairer Access and Lower Losses

Data-driven decisions can reduce defaults while expanding responsible credit.



Key Insights from the Portfolio

What the Data Tells Us

Renters and Unemployed Borrowers Drive Default Risk

Borrowers without housing stability or full-time employment represent the highest-risk segment in Nova Bank's portfolio. Their elevated debt-to-income ratios and financial volatility demand stricter affordability checks and revised approval thresholds.

Education and Venture Loans Deliver Strongest Repayment Performance

Borrowers seeking education or launching ventures show the most reliable repayment behavior. These segments align with long-term income growth and should be prioritized in product design and interest rate strategy.

Younger Applicants Pose Elevated Risk Due to Credit Immaturity

Youth and young adults (18–35) dominate loan applications but carry higher default rates due to short credit histories and high loan-to-income ratios. Nova Bank should apply tighter checks and offer credit-building pathways for this group.

Grade D–G Loans Are High-Risk — Especially at Long Terms

Default rates exceed 60% for Grade D loans and reach 100% for Grade G loans at 60-month terms. These products expose Nova Bank to unsustainable risk and should be capped or restructured with stricter eligibility criteria.

01

Youth (18–25)

High loan-to-income ratio,
short credit history

02

Renters

Highest debt and default
volume

03

Grades D–G

Up to 100% default rate at
long terms

04

**Employment <2
years or >20 years**

elevated risk

Who Defaults and Why

Risk Profiles That Matter



What Nova Bank Can Do

Lending Strategy

Optimize Lending Around Proven Risk Indicators

Prioritize Grade A–C loans and cap risky loans (Grades D–G) at 36 months. These grades consistently show lower default rates and stronger repayment behavior, especially at shorter terms. This protects profitability while maintaining access.

Refine Risk Scoring with Behavioral and Intent-Based Features

Incorporate loan intent, credit history length, and account behavior into scoring models. These features improve early risk detection and align with the objective to identify borrower groups more or less likely to default.

Apply Stricter Affordability Checks for High-Risk Segments

Tighten approval criteria for renters, unemployed applicants, and youth segments. Use loan-to-income and debt-to-income ratios as gating thresholds to flag over-leveraged borrowers and reduce exposure in high-risk regions.

Empower Borrowers Through Financial Literacy and Product Design

Offer financial literacy tools and starter loan products for vulnerable groups. This supports Nova Bank's goal of fair lending while improving repayment outcomes and long-term customer value.

Conclusion

Smarter Lending Starts with Smarter Data

- Nova Bank's credit risk profile reveals clear patterns in borrower behavior, loan structure, and default exposure
- By aligning lending policies with these insights, the bank can reduce losses while expanding responsible access to credit
- These recommendations support smarter lending decisions that protect the portfolio and empower borrowers
- The next step: embed these strategies into scoring models, product design, and policy frameworks